Arithmetic, Coverings and Learned Quantum Renormalization

Resource-certified elliptic circuits, gentle orders, hierarchical channels and observed CMB controls

Research continuation prepared for Amelie Schreiber

14 November 2026

Program-three source and execution edition

The implemented continuation

The subsection entitled *Resource-certified elliptic-curve quantum arithmetic* asked for a circuit generator, an explicit reversible specification, state-vector checks, independent resource accounting and a machine-readable assumption record. This continuation supplies those artifacts for completely enumerated finite instances. It also develops the proposed connection to covering theory and learned quantum coarse graining, then applies the resulting hierarchical channel to a frozen broad-feature search on actual WMAP maps.

The unifying object is an explicitly represented map with a checkable contract. Arithmetic maps must be permutations on their declared input space; covering maps must preserve stars and relations; coarse-graining maps must be completely positive and trace-preserving; learned search statistics must be frozen before the observed score is inspected. These are different contracts, linked by concrete constructions rather than by notation alone.

The historical 61-page *Certified Surface Computation* volume and its evidence remain unchanged. This is its next implemented research layer. The exact source, emitted circuits, learned weights, observations, conditional null simulations, exploratory followups, formal logs and interactive views accompany the publication. A GPU simulation is recorded as such, and a conditional observational rank retains the null hypothesis that gives it meaning.

Contents

1 Resource-certified elliptic arithmetic **1**
- 1.1 A complete specification before a resource claim . . . 1
- 1.2 An emitted gate construction with clean work . . . 1
- 1.3 Field and variable-point oracles . . . 2
- 1.4 A fully paid endomorphism comparison . . . 2
- 1.5 Formal and numerical verification layers . . . 3

2 Coverings, orders and learned quantum renormalization **4**
- 2.1 Voltage covers preserve the specified local presentation . . . 4
- 2.2 Integral relations can obstruct a special-fiber cover . . . 5
- 2.3 Neural pullback, descent and the cost of arbitrary identifiers . . . 5
- 2.4 A learned hierarchy of physical quantum channels . . . 6
- 2.5 What the learned renormalization map actually learns . . . 6
- 2.6 An explicit infinite-width theorem . . . 7
- 2.7 Cover symmetry and the dependence structure of wide quantum networks . . . 8
- 2.8 Testing whether gate learning improves the representation . . . 9

3 CCC geometry and a real-map search with frozen learned statistics **10**
- 3.1 The original mathematical interfaces are retained . . . 10
- 3.2 Actual datasets and resolution-aware tests . . . 10
- 3.3 Learned statistics selected before observation scoring . . . 11
- 3.4 The null is part of the result . . . 11
- 3.5 Observed results and frequency sensitivity . . . 12
- 3.6 Interactive replay without moving the goalposts . . . 13
- 3.7 A joint phase action that retains cross-product dependence . . . 13

4 Execution and reproducibility **14**

A Primary sources and evidence boundaries **18**

Chapter 1

Resource-certified elliptic arithmetic

1.1 A complete specification before a resource claim

The original section's criterion is a comparison between complete resource budgets. The implementation fixes a nonsingular short-Weierstrass curve $E : y^2 = x^3 + ax + b$ over a declared odd prime field, a public fixed generator G, an affine coordinate encoding, a gate alphabet and an ancillary-register policy. It generates ordinary fixed-base multiplication and a signed endomorphism decomposition under the same compiler. A shorter arithmetic expression is not counted as an improvement until every operation needed to realize it has been charged.

For $b_p = \lceil \log_2 p \rceil$, the point register has $2b_p + 1$ bits: b_p each for x, y, and a unique infinity word with the top bit set and both coordinates zero. A valid finite point has top bit zero, coordinates below p, and satisfies the curve equation. Every other bit string is invalid. Each in-place point translation acts as the group translation on valid words and fixes every invalid word. This extends the mathematical permutation to a total reversible operation on the register.

Proposition 1.1 (Total translation and coherent extension)**.** *For fixed $Q \in E(\mathbb{F}_p)$, the above map is a permutation of all point words. Its inverse is the extension of translation by $-Q$. Its permutation matrix is unitary, with no input-dependent phase.*

Proof. The valid words and invalid words are disjoint invariant subsets. Translation is a bijection on the first with inverse translation by $-Q$, and the identity is a bijection on the second. Their disjoint union is a permutation. Its columns are distinct standard basis vectors and therefore form an orthonormal basis. Linear extension establishes the action on every superposition. □

The scalar circuit acts on an arbitrary initial valid point, not only infinity:

$$|k, P, 0\rangle \longmapsto |k, P + kG, 0\rangle.$$

It preserves the scalar register and restores all declared scratch and decomposition registers. Invalid P remains fixed. The output is deterministic under the ideal logical-gate model.

1.2 An emitted gate construction with clean work

Every finite permutation is decomposed into cycles. A cycle is realized by a sequence of transpositions between one anchor and the remaining cycle entries. To swap two bit strings, choose a Gray path between them. Apply its adjacent transpositions in forward order and then reverse order omitting

the last transposition. The resulting permutation swaps only the endpoints; the internal path entries return to their initial locations.

An adjacent Gray transposition flips one target bit conditional on all other data bits matching a pattern. Negative controls are implemented by an X conjugation on the corresponding control wire. A k-controlled NOT with $k \geq 3$ is lowered to an AND ladder, one final controlled flip, and the reverse ladder. It uses $k-2$ clean work bits and $2k-3$ CCX gates. For zero, one or two controls the primitive is X, CX or CCX directly.

Proposition 1.2 (Clean controlled flip)**.** *On inputs whose ladder work bits are zero, the emitted ladder toggles the target exactly when every control is one, leaves the controls unchanged and restores every work bit to zero.*

Proof. The first CCX stores the conjunction of the first two controls. Inductively each following ladder bit stores the conjunction of the controls accumulated so far. The last CCX therefore toggles the target exactly under the full conjunction. The target is not a control in the ladder. Reversing the ladder toggles each stored conjunction off in reverse dependency order while its input controls still have their original values. Negative-control conjugation changes only which input pattern selects the flip and is undone afterward. □

The compiler emits every primitive rather than treating the controlled operation as a free black box. An independent parser counts the serialized primitives and reconstructs their all-to-all dependency depth. This count is separate from the generator's own count. Physical routing, fault tolerance, logical-error targets and T-gate synthesis are not inferred from an X/CX/CCX table. Those require another explicitly selected compilation architecture.

1.3 Field and variable-point oracles

The field reference supplies XOR-output addition, multiplication and inversion. For inversion, zero has a separate false validity flag rather than an invented inverse. Invalid field words contribute zero oracle output. The general reversible specification is

$$|x, z, 0\rangle \longmapsto |x, z \operatorname{xor} f(x), 0\rangle.$$

Because XOR by a fixed word is an involution for each retained x, arbitrary initial output words are supported.

The variable-point addition oracle retains both input points and writes the valid result together with a validity bit. Invalid point input pairs contribute zero to the output register. All valid pairs, all invalid input encodings, and independent nonzero output words are tested. This complements the in-place fixed-base translation circuit; it is not presented as the same register layout or the same resource budget.

The lookup/Gray-path compiler is deliberately explicit and bounded. Its worst-case table cost grows exponentially in the bit width. This is a complete finite reproducibility reference and a test bed for replacing each primitive by a scalable field-arithmetic circuit. It is not a claim of having improved cryptographic-size resource estimates such as [E1]. A future efficient replacement must pass the same total-domain and cleanup contracts before its resource table is comparable.

1.4 A fully paid endomorphism comparison

For the tested $j = 0$ curves, choose a nontrivial cube root of unity $\beta \in \mathbb{F}_p$ and define

$$\phi(x, y) = (\beta x, y), \qquad \phi(O) = O.$$

The generator's cyclic subgroup has $\phi(G) = [\lambda]G$. The finite decomposition table chooses signed pairs (a_k, b_k) satisfying

$$k \equiv a_k + \lambda b_k \pmod{\text{ord}(G)}.$$

The circuit coherently computes the signed pair into clean registers, performs controlled additions of the signed powers of G and $\phi(G)$, and uncomputes the pair. Two's-complement sign bits use negative powers in their corresponding controlled addition. Every scalar register word is interpreted modulo the declared subgroup order.

Here the base is public and fixed, so $\phi(G)$ is an explicitly recorded classical constant. This does not make evaluation of $\phi(P)$ on an unknown variable point free. Nor does a fixed-base table justify replacing a general lattice-decomposition algorithm by a costless oracle. The current comparison pays for the exact finite decomposition it actually executes.

Field	Path	Qubits	X	CCX	Depth
7	binary	16	2328	2904	3390
7	endomorphism	22	3808	4766	5608
13	binary	21	7040	8850	9832
13	endomorphism	27	9204	11768	13186

The endomorphism path costs more in both measured instances. For $p = 7$, it uses 4,766 rather than 2,904 CCX gates; for $p = 13$, it uses 11,768 rather than 8,850. The complete register counts rise from 16 to 22 and from 21 to 27, respectively. These are decisive answers for this compiler and these instances. They neither rule out a better compiler nor establish a saving at a different field size.

1.5 Formal and numerical verification layers

The Python suite checks the group operation on the complete small point sets, every declared scalar/point data word, invalid encodings, reverse-circuit replay and clean work. The $p = 7$ fixed-base domain contains 2,048 scalar/point words; the $p = 13$ domain contains 16,384. The gates are phase-free permutations, so complete basis semantics fixes their coherent linear extension. The separately executed GPU test verifies that extension on complex amplitudes across the entire $p = 7$ data domain.

The generated Lean certificate uses a direct modular affine-addition formula to check the emitted small-curve table, then checks all table associativity, identity, inverse, endomorphism and scalar-decomposition instances. These are exact finite propositions elaborated by the kernel, not statistical tests. General cleanup and normalized-map lemmas are formalized separately. Neither an exhaustive finite table nor a successful GPU comparison is represented as a formal asymptotic complexity theorem.

This completes the original subsection's concrete artifact program at its declared finite scope: a generator, reversible specification, total-domain tests, actual state-vector execution, independent gate counts and an assumption manifest. The next optimization question now has a reproducible baseline against which a claimed improvement can be falsified.

Chapter 2

Coverings, orders and learned quantum renormalization

2.1 Voltage covers preserve the specified local presentation

Let Q be a finite quiver and assign each arrow $a : u \to v$ a voltage $\nu(a)$ in the cyclic group $\mathbb{Z}/d\mathbb{Z}$. The lift has vertices (u, s) and arrows

$$(a, s) : (u, s) \longrightarrow (v, s + \nu(a)).$$

Translation of every sheet index by the same element is a deck action. A path lifts uniquely after its initial sheet is supplied. A monomial zero relation lifts to the corresponding zero relation on each lifted path.

Proposition 2.1 (Finite gentle cover)**.** *If a finite-dimensional quadratic monomial bound quiver is gentle, the voltage lift with the lifted monomial relations is gentle and finite-dimensional.*

Proof. At each lifted vertex, incoming and outgoing arrows correspond bijectively to those at the base vertex. Thus the degree-at-most-two conditions are unchanged. For each lifted arrow, allowed and forbidden successor and predecessor choices correspond bijectively to the base choices, so their uniqueness conditions persist. Any allowed lifted path projects to an allowed base path of the same length. Since the base's allowed-path lengths are bounded, the lifted allowed-path lengths are bounded as well. For a finite quadratic monomial presentation this gives finite dimensionality. □

The implementation checks every voltage assignment in selected small covers, not only one favorable drawing. It also constructs explicit unramified rectangular-torus cell covers, checks both cellular coboundary pullback identities and verifies $d_1 d_2 = 0$. The degree-d torus cover still has genus one. This is consistent with multiplicativity of Euler characteristic for an unramified cover, not evidence that an arbitrary quiver cover automatically extends to a marked, branched or holomorphic surface cover.

For a surface with boundary, marked points or ramification, the covering data must include those structures. The gentle algebras associated with unpunctured marked surfaces in [G1] have specific hypotheses. A general categorical covering functor [G2], a voltage cover of a presentation, and a topological covering of a dessin surface are related constructions but not interchangeable definitions.

2.2 Integral relations can obstruct a special-fiber cover

The rank-four order of the preceding volume has generators satisfying

$$uv = \pi e_1, \qquad vu = \pi e_2.$$

A lift that assigns independent sheet shifts to u and v makes their composite act by the total sheet shift. Over a torsion-free module with nonzero π, preserving the displayed relation requires that total shift to be the identity. In the explicit matrix realization the residual is

$$\pi(S_{\nu(u)+\nu(v)} - I).$$

Reduction modulo π turns the relation into a zero relation. The special fiber therefore admits covers that fail the original integral relation. The code checks this obstruction at degree two: voltages $(1, 0)$ fail at $\pi = 3$ but pass after specialization to zero; voltages $(1, 1)$ preserve the integral relation. In this two-arrow example, zero cycle voltage also makes the resulting cover split into components. A connected nontrivial special-fiber cover is not thereby a connected cover of the same integral order.

More general order coverings require a declared grading, central-parameter action, local lattice data and compatibility with all nonmonomial relations. The present obstruction gives an exact test to apply before extending a neural or categorical construction over the generic fiber. Forgetting the deformation parameter can genuinely remove information needed to validate the lift.

2.3 Neural pullback, descent and the cost of arbitrary identifiers

For a d-sheet finite cover let U pull a base feature to a constant feature on each fiber. Over $\mathbb{R}$, put $P = d^{-1}U^{\mathsf{T}}$. Then

$$PU = I, \qquad (UP)^2 = UP.$$

The second map projects onto fiber-constant signals. In characteristic dividing d, this average is not defined by division; replacing it by an unnormalized sum can even send the constant-one feature to zero. The implementation retains that counterexample.

Proposition 2.2 (Local layer transport)**.** *Suppose a base message-passing update uses a shared message function, a shared node update, and a sum or normalized local mean over incoming arrows. On pullback features, the same update on the cover commutes with U.*

Proof. Every lifted vertex has one incoming lift of each incoming base arrow. The message arguments are equal because the source and target features are pulled back. Therefore the local sum is the base sum; the local degree is also the same, so normalized means agree. Applying the common node function preserves equality. Induction gives the result for every finite composition of such layers. □

There is a related global-attention identity. If each key/value is replicated exactly d times, then each softmax numerator and its denominator receive the same factor d, leaving the weighted value unchanged. Repetition also leaves a coordinatewise maximum unchanged. Consequently normalized softmax and max-plus operations commute with the declared uniform replication. This statement requires transported or tied features and parameters. Arbitrary absolute sheet identifiers break it, as the negative test demonstrates. Global attention on arbitrary non-fiber-constant states is not covered by the identity.

The normalized linear embedding $J = U/\sqrt{d}$ is an isometry. It is a direct-sum encoding of amplitudes, not a map that clones an unknown quantum state into d independent tensor copies. That

distinction preserves the no-cloning constraints already proved in the CCC chapter. A quantum circuit descends through a cover only after its state embedding and operator intertwining are specified.

Covering alone also does not preserve a positive lower spectral gap. A connected d-fold cover of a finite cycle is a longer cycle; its smallest positive graph-Laplacian eigenvalue is $4\sin^2(\pi/(dN))$, which tends to zero. Base eigenvectors lift, but new non-fiber-constant modes appear. Thus cover degree is a geometric limit requiring its own estimates, not a substitute for the infinite-width limit of a neural parameterization.

2.4 A learned hierarchy of physical quantum channels

The implemented network coarse grains four qubits to two and then one. Each stage applies explicitly specified two-qubit unitary blocks and traces out the declared environment qubits. The first stage shares three rotation parameters across its two pairs, and the second stage uses another three. A scalar readout adds slope and bias; it is a classical postprocessor, not part of the quantum channel.

For a stage unitary W and an environment basis $\{|e\rangle\}$, the Kraus operators are

$$K_e = (I \otimes \langle e|)W.$$

Their completeness follows directly:

$$\sum_e K_e^* K_e = W^* \left(I \otimes \sum_e |e\rangle\langle e| \right) W = I.$$

If L_f describes the next complete stage, the combined operators are $L_f K_e$ and

$$\sum_{e,f} (L_f K_e)^* (L_f K_e) = \sum_e K_e^* \left(\sum_f L_f^* L_f \right) K_e = I.$$

The general rectangular-matrix identity is checked in Lean. An independent numerical implementation constructs the Kraus operators, while the training path uses a Heisenberg observable and another reference performs explicit partial traces. Their agreement checks both wire order and discarded-subsystem semantics.

A discarded qubit need not be physically erased during a simulation. If no later gate acts on it, postponing its trace to the end gives the same retained density matrix. This allows the complete four-qubit unitary circuit to be replayed independently on CPU and cuStateVec. It does not postselect favorable outcomes, and it does not replace a channel by a nonlinear normalization of a selected branch.

2.5 What the learned renormalization map actually learns

Training uses exact Gibbs states of a four-site periodic transverse-field Ising Hamiltonian at inverse temperature one. The 33 coupling values are split by a declared index rule into train, validation and test. The objective retains the mean nearest-neighbor ZZ observable at the final readout. The six gate angles and two readout parameters are actually optimized; validation selects the checkpoint and test couplings do not select it.

Seed	Selected step	Test MSE	Constant baseline
17	120	0.0000615	0.0260293
29	120	0.0009693	0.0260293

This is a learned coarse-graining channel with a measurable target, not an assertion that every relevant operator is preserved. The complete first-stage two-qubit density matrix is retained, and its effective Hamiltonian is reconstructed as

$$H_{\text{eff}} = -\log \rho_{\text{coarse}} + cI.$$

Every finite-temperature input is positive definite. Unitary conjugation preserves that property, and a partial trace of a positive-definite operator is positive definite: for nonzero v, sum the strictly positive quadratic forms on $v \otimes e$. Thus the matrix logarithm is defined for these exact states.

The implementation expands H_{eff} in all sixteen two-qubit Pauli operators. Projecting it back to only IX, XI and ZZ is a separate model reduction. The trace distance between the actual coarse state and the Gibbs state of that truncated Hamiltonian is reported at every coupling. Nonzero generated interactions are not discarded while claiming exact closure within the original Ising family.

A useful stability statement follows from the integral formula for the matrix logarithm. If positive definite A, B have minimum eigenvalue at least $\mu > 0$, the resolvent identity gives

$$\|\log A - \log B\| \le \|A - B\| \int_0^\infty (\mu + t)^{-2}\, dt = \frac{\|A - B\|}{\mu}.$$

A learned coarse state close to a singular state can therefore have an unstable effective Hamiltonian even when its density-matrix error is small. The recorded minimum eigenvalues expose this condition instead of hiding it in a renormalization plot. An infinite-system fixed point, scaling exponent or universal phase transition needs a system-size analysis beyond these four-site experiments.

2.6 An explicit infinite-width theorem

The width study uses independent one-qubit feature circuits, not the number of sheets in a cover. Let $w_r \sim N(0, I_d)$ and $b_r \sim \text{Unif}[0, 2\pi]$ independently, and define

$$\varphi_r(x) = \sqrt{2}\, \langle Z \rangle_{R_Y(w_r \cdot x + b_r)|0\rangle} = \sqrt{2}\cos(w_r \cdot x + b_r).$$

Set $K_M(x, y) = M^{-1} \sum_{r=1}^{M} \varphi_r(x)\varphi_r(y)$.

Theorem 2.3 (Independent quantum features and their limiting kernel)**.** *For fixed x, y,*

$$K_M(x, y) \longrightarrow K(x, y) = \exp(-\|x - y\|^2/2)$$

almost surely. On a fixed set of n inputs,

$$\Pr\left(\max_{i,j} |K_M(x_i, x_j) - K(x_i, x_j)| > \epsilon \right) \le 2n^2 e^{-M\epsilon^2/8}.$$

Every empirical kernel is positive semidefinite.

Proof. Averaging over b gives $\mathbb{E}_b[2\cos(a+b)\cos(c+b)] = \cos(a-c)$. The characteristic function of a standard Gaussian then gives the displayed K. The strong law applies to the independent bounded products. Each product lies in $[-2, 2]$, so Hoeffding's inequality gives $2e^{-M\epsilon^2/8}$ for a fixed pair. A union bound over the n^2 pairs proves the finite-set estimate. Positive semidefiniteness follows because K_M is a feature Gram matrix. □

The empirical study uses widths 16, 64, 256, 1,024 and 4,096 and four fixed seeds, preserving all errors rather than requiring every realization to improve monotonically. The random-feature circuit is classically tractable; the example establishes a controlled quantum-feature limit, not a computational speedup.

Readout-only kernel ridge regression has a corresponding stability estimate. If $\|K_M - K\|_2 \le \delta$ and regularization is λI with $\lambda > 0$, the resolvent identity gives

$$\|(K_M + \lambda I)^{-1} y - (K + \lambda I)^{-1} y\|_2 \le \delta \|y\|_2 / \lambda^2.$$

This is a specific trained readout regime. It does not prove that training every gate of an unrestricted deep quantum network leaves the kernel fixed.

The trained-network Gaussian-process results in [Q2] are relevant to broader architectures, but their initialization, light-cone dependence, normalization, limiting kernel and learning-rate assumptions must travel with the theorem. A deep hierarchy with expanding causal cones, shared parameters or max-plus branches does not inherit the result merely by increasing a register count. Cover limits, feature-width limits, physical thermodynamic limits and optimizer-time limits remain four distinct operations.

2.7 Cover symmetry and the dependence structure of wide quantum networks

Exact symmetry and independent random features need not be compatible under an arbitrary parameter-sharing scheme. If every sheet uses the same circuit parameters, repeating a feature does not create independent samples. In particular, for one random feature φ, the empirical expression $M^{-1} \sum_{r=1}^{M} \varphi(x)\varphi(y)$ is just $\varphi(x)\varphi(y)$ for every M. It does not converge to the averaged RBF kernel by increasing the number of identical copies. The additional executable control retains this rank-one counterexample alongside a genuinely nested sequence of independent features.

A useful general criterion can nevertheless be stated in terms of light-cone dependence. Consider centered random circuit readouts $X_{M,1}, \dots, X_{M,M}$, uniformly bounded by a constant, with a dependency graph of maximum degree D_M. This means that two collections with no edge between them are independent. Suppose $M^{-1} \operatorname{Var}(\sum_r X_{M,r}) \to \sigma^2$ and $D_M = o(M^{1/4})$.

Proposition 2.4 (A sufficient Gaussian finite-dimensional limit)**.** *Under these assumptions, $M^{-1/2} \sum_r X_{M,r}$ converges in distribution to $N(0, \sigma^2)$. The same assertion holds jointly on any fixed finite input set when the bound and dependency graph apply to the corresponding readout vectors and their normalized covariance matrices converge.*

Proof. For each fixed order k, the cumulant of the sum is a sum of joint cumulants. A joint cumulant vanishes when its variables split into independent groups. Thus only tuples connected in the dependency graph, allowing repeated indices, contribute. There are at most $C_k M (D_M + 1)^{k-1}$ such tuples. Boundedness gives a uniform fixed-order cumulant bound by the moment–cumulant formula. After normalization, the order-k bound is therefore $C'_k M^{1-k/2} (D_M + 1)^{k-1}$.

For $k = 3$ this tends to zero by $D_M = o(M^{1/4})$; for every fixed $k > 3$ it also tends to zero. The first cumulant is zero and the second converges by assumption. The moment–cumulant relations therefore give convergence of all moments to the Gaussian moments. The Gaussian distribution is moment-determinate, so the method of moments gives convergence in distribution. For a finite vector of inputs, apply the same argument to each fixed linear combination and use the Cramér–Wold criterion. □

For randomly initialized circuits, disjoint parameter light cones can supply the required dependency graph. A global shared parameter or a growing cone can violate the bound. Training all gates with a common data-dependent loss can introduce new dependencies, so the initialization theorem is not automatically a theorem about the trained network. The trained-QNN results in [Q2] address this transition under their additional kernel, light-cone and scaling assumptions.

One symmetry-compatible construction uses independent feature channels, each of which is averaged over the same finite deck group. Every channel is then exactly equivariant or invariant as specified, while different channels remain independent. The width limit is taken over those independently initialized channels on a fixed covered graph. Increasing the covering degree is a different limit and can change both the dependency bound and the spectrum. This provides an explicit design rule rather than conflating replication, geometric lifting and statistical width.

2.8 Testing whether gate learning improves the representation

A low error relative to a constant predictor does not by itself show that optimizing quantum gates was necessary. The additional control freezes each seeded initial channel, fits only its scalar affine readout on training couplings, and evaluates the same held-out couplings. It reports the training design's condition number, the frozen-channel/readout error, the raw initialized error and the fully learned channel's error. No learned weights are retrained or selected using this comparison.

The full density matrix and the declared ZZ observable also supply an exact target oracle. Its zero error is a statement about access to the uncompressed input, not an equally constrained learned baseline. Together these comparisons distinguish three effects: information already present in the input, information retained by the channel, and information made accessible by its trained readout. The publication retains a failure to improve over a frozen channel whenever it occurs.

Seed	Learned gates MSE	Frozen gates + readout	Constant MSE
17	0.0000615	0.0001169	0.0260293
29	0.0009693	0.0000056	0.0260293

Chapter 3

CCC geometry and a real-map search with frozen learned statistics

3.1 The original mathematical interfaces are retained

The original CCC chapter separates conformal geometry, elliptic estimates, quantum-information constraints and observational calibration. This continuation strengthens those interfaces instead of replacing them by a generic neural classifier. The formal additions check the middle-form conformal exponent and the explicit polynomial transformations used by the elliptic-curve discussion. They do not turn an algebraic identity into an observational confirmation of a crossover.

For a metric rescaling in dimension n, the Hodge-star scaling on k-forms has exponent $n-2k$. In four dimensions at degree two this exponent is zero. The new finite identity checks that exponent's specialization. The full Hodge-star transformation law, metric signatures and Maxwell/source assumptions remain the mathematical premises stated in the original chapter.

Let $P(x) = ax^4+bx^3+cx^2+dx+e$ and let r be a root. Substituting $x = r+1/u$ and multiplying by u^4 produces

$$u^4P(r+1/u) = Au^3 + Bu^2 + Cu + D,$$

where

$$A = P'(r), \quad B = P''(r)/2, \quad C = P'''(r)/6, \quad D = P''''(r)/24.$$

For $v^2 = Au^3 + Bu^2 + Cu + D$, put $X = Au + B/3$ and $Y = Av$. Then

$$Y^2 = X^3 + (AC - B^2/3)X + 2B^3/27 - ABC/3 + A^2D.$$

Lean checks the coefficient expansions and transport identity. A birational interpretation additionally requires the stated nonzero denominators and a simple root when A is used to invert the coordinate change. Nonsingularity is a separate discriminant condition. The same algebraic shape across finite-field arithmetic and conformal-time equations is useful structure, not an identification of their physical interpretations.

3.2 Actual datasets and resolution-aware tests

The study uses three actual WMAP nine-year products: ILC and foreground-reduced V- and W-band temperature maps, plus the KQ85 mask. All are acquired through links on official NASA LAMBDA product pages. The acquisition record binds filenames, source URLs, exact bytes, FITS fields, native Nside512 nested ordering and temperature units. Millikelvin are explicitly converted

to microkelvin; the ILC header's `mK, thermodynamic` wording is retained rather than rejected or silently normalized in the source record.

The ILC documentation states reliability above approximately ten degrees and cautions about uncertainty at smaller angular scales [C1]. Accordingly this implementation tests broad Gaussian widths of 12, 18 and 24 degrees. It is not presented as a reproduction of a degree-scale Hawking-point search. Twenty-four fixed high-latitude centers and three scales give 72 templates.

Processing is fixed: pixel-average to Nside16, estimate harmonics through $\ell = 32$, remove monopole and dipole, apply an additional five-degree Gaussian beam, and reconstruct the map. Retained pixels require degraded KQ85 fraction at least 0.8 and absolute Galactic latitude at least twenty degrees. The original map beams differ; adding the same smoothing does not constitute exact common-beam deconvolution. The processing record retains that fact.

Three statistics are scanned. The first compares a Gaussian-weighted temperature mean with the retained-sky mean. The second detects low variance in the outer radial annulus. The third applies a learned four-qubit hierarchy to four mean-subtracted radial summaries. The corresponding local graph is an ordered four-band relation, not a privileged target-derived topology.

3.3 Learned statistics selected before observation scoring

The channel architecture is the same explicit 4-to-2-to-1 construction used in the preceding chapter. Its gate angles and scalar readout are trained on independently generated phase-surrogate skies and injected broad profiles. The observed map's original phases do not select the weights, center grid, scale set, mask, amplitude grid or checkpoint.

There are 1,024 training examples and 256 held-out validation examples, with Gaussian-profile amplitudes 64, 128 and 256 microkelvin. A bounded arctangent encoding maps four radial features to rotation angles, with the scale estimated from training only. Validation selects among 150 optimizer steps. The code freezes the selected gates, input scale and readout before computing the observed full-search scores.

This is an actually trained, classically simulated QNN, not quantum hardware and not a trained full Mora-3 model. Its classifier is a candidate statistic for a specified injected alternative. Success on that injection task does not establish that every physical CCC signal has that form, or that another physical process cannot produce similar features.

3.4 The null is part of the result

For each processed map, null surrogates multiply $m > 0$ harmonic coefficients by independent uniform phases and $m = 0$ coefficients by independent random signs. Every coefficient magnitude is retained. Monopole and dipole remain absent. Training, reference-normalization, calibration and validation use distinct seeds and realizations.

This conditional phase-randomization null is scientifically inspectable, but it is not a complete WMAP likelihood. Anisotropic instrumental noise, foreground-cleaning uncertainty, harmonic estimation and masking effects are not fully modeled. Exact randomization validity requires invariance under the declared phase action. A small rank can therefore diagnose disagreement with this surrogate law without uniquely identifying CCC.

For every calibration sky, the pipeline recomputes the same location, scale and method maximum as for the observed sky. There are 511 calibration draws, so the smallest reported rank is 1/512, never zero. Independent null validation uses 256 draws and reports its rejection fraction and Wilson

interval. Injection sensitivity uses 64 additional controls per declared amplitude, including zero amplitude. All products and all outcomes remain in the record.

Theorem 3.1 (Conservative finite upper ranks, including ties)**.** *For n real scores s_i, define $r_i = \#\{j : s_j \geq s_i\}$. Then for every integer k,*

$$\#\{i : r_i \leq k\} \leq k.$$

If the distinguished index is uniform under exchangeability, this gives the usual conservative finite-rank probability bound.

Proof. If the set on the left is empty there is nothing to prove. Otherwise choose a member with minimum score among its members. Every member's score is at least that minimum, so the whole set is included in the chosen member's upper-rank set. Its cardinality is at most that upper rank, which is at most k. Ties are included in the upper-rank set and require no arbitrary tie-breaking assumption. Uniformity of the distinguished index then divides the count by n. □

The deterministic counting theorem is formalized for arbitrary finite score vectors, not only one simulated sample. Exchangeability of real observations and a physical null is an additional modeling premise. The earlier measure-theoretic family bounds remain applicable once their event-level premises are justified.

3.5 Observed results and frequency sensitivity

Product	Full-search rank	Three-product bound	Null rejections
ILC	0.992188	1.000000	15/256
V	0.005859	0.017578	16/256
W	0.238281	0.714844	11/256

The original workstation run found a small joint phase-surrogate rank for the V-band product, but not for ILC or W-band. The independent runner preserves its own fully recorded recomputation rather than replacing it with those earlier numbers. These products share the CMB sky and observations; their ranks must not be multiplied as though they were independent confirmations. The displayed three-product correction is a union-bound allocation, not evidence that the underlying surrogate model is physically complete.

The search retains the selected template, per-method maxima and all calibration maxima. The low-variance statistic can be unusual under its own method-specific null while the joint maximum has a different tail. This is why both are reported and why the local selected tail is not substituted for the full-search result.

After the initial product disagreement was inspected, a separate exploratory followup was designed. It applies a common ILC-derived phase reference to all products and scans the processed V-minus-W map. It also reports masked cross-product correlations and map amplitudes. This followup is explicitly marked post-inspection. It does not replace the predeclared scan, is not added as independent evidence, and is not folded into a retroactively enlarged family with an unchanged significance claim.

The V-minus-W map is a sensitivity diagnostic, not a pure foreground estimate: different effective beams, noise and cleaning residuals remain. Similarly, changing from each product's conditional harmonic magnitudes to a common reference changes the null hypothesis. Those changes are exactly what a robustness analysis should expose, rather than conceal behind a single favorable number.

3.6 Interactive replay without moving the goalposts

The sky view displays observed temperatures on a Galactic Mollweide projection, with the fixed exclusion mask visibly separated. Product, statistic, center and injected amplitude are controlled independently. At zero amplitude the original observed map is replayed. At nonzero amplitude the display says that a synthetic profile has been added. All 72 templates and all three frozen statistics are then recomputed with unchanged weights and unchanged calibration.

The browser contains its own numerical implementation, including the actual four-qubit gates. It is compared with Python on all observed products and additional injected cases. A source-hash mismatch prevents the page from presenting an unverified data object as an accepted result. Exported state records the source commit, asset hashes, arithmetic outputs, quantum readouts and complete-search rank.

This completes a concrete CCC-oriented observational extension: real source products, a frozen learned statistic, full-search controls, independent null validation, injection sensitivity, explicitly exploratory frequency tests, formal rank logic and an interactive replay. It does not claim to have established a physical cross-aeon channel or an intentional message. Those questions require alternative source models with distinguishable predictive consequences, as the original information-theoretic chapter explains.

3.7 A joint phase action that retains cross-product dependence

A stronger dependence-aware control acts on the three harmonic maps together. For each mode choose one common unit-modulus phase $\zeta_{\ell m}$ and transform every product by

$$a^{(A)}_{\ell m} \longmapsto \zeta_{\ell m} a^{(A)}_{\ell m}.$$

Use a real sign for $m = 0$. Then every complex harmonic cross-product is preserved exactly in arithmetic:

$$(\zeta a^{(A)})\overline{(\zeta a^{(B)})} = a^{(A)}\overline{a^{(B)}}.$$

The implementation checks this identity numerically for every surrogate draw. Independent phases per product would preserve auto-power but generally destroy these cross-products; a negative test demonstrates the difference.

The new statistic is one maximum over all three products, all three methods and all 72 templates. Reference normalization, 511 calibration draws and 256 independent validation draws use the shared phase action. This gives one conditional family-wide rank without assuming independence of the product maps and without multiplying their ranks. The learned gates are unchanged.

This control was designed after the first product-specific scan and is labeled exploratory. It improves the representation of cross-product dependence, but it does not turn phase-group invariance into an established fact about the complete instrument and foreground-processing model. Its interpretation is still conditional on that hypothesis. Both the original scan and the new family control are retained, including disagreement between them.

Shared-phase family	Joint rank	Null rejections
V selected	0.017578	9/256

The common-reference, frequency-difference and joint-phase tests therefore answer different robustness questions. None is selected retrospectively as the uniquely valid answer because it gives a preferred tail. Their frozen configurations, source hashes, score arrays and validation outcomes make those distinctions reproducible.

Chapter 4

Execution and reproducibility

The independent runner accepted 22 new formal declarations, 142 regression methods under each Python mode, 32 browser checks, and 9 complete JavaScript/Python sky-score replays. The maximum score discrepancy was 3.69e-14.

The published Python tests exercise the ordinary and optimized interpreters. The formal audit prints each new declaration and its axiom inventory, rejects warnings, rejects `sorryAx`, and binds the exact source to Lean 4.24.0 and the recorded mathlib revision. The finite arithmetic module checks its entire small curve table; the general Kraus and rank-count modules verify their displayed quantified statements. The whole physical interpretation is not encoded by a successful process exit.

The browser executes the emitted reversible gates and the learned four-qubit circuit. It independently recomputes all 72 fixed sky templates and all three methods for observed maps or explicitly labeled injected profiles. Its scientific JSON assets are SHA-256 checked before use. Interactive state is exportable and URL-backed; keyboard and mobile behavior are tested. A corrupted asset is rejected rather than displayed under a successful-verification badge.

The original workstation completed genuine cuStateVec checks of both $p = 7$ arithmetic circuits on a complex superposition spanning all 2,048 declared input words. The observed maximum amplitude discrepancy and dirty-ancilla probability were zero. Six trained-channel gate checks agreed with the independent Kraus implementation to 5.56×10^{-16}. The subsequent independent repository runner has no GPU and does not relabel those measurements as its own. The $p = 13$ circuits retain exhaustive basis verification without a dense-GPU claim.

Desktop Commander reached its monthly usage limit during the continuation. No retry, reconnection, service restart, credential extraction or alternate access to the paused device was used. Independent repository CI rebuilt the source and evidence on a standard public CPU runner. Remote clone synchronization after that pause is not asserted. The original Mora-3 training campaign remains outside this task.

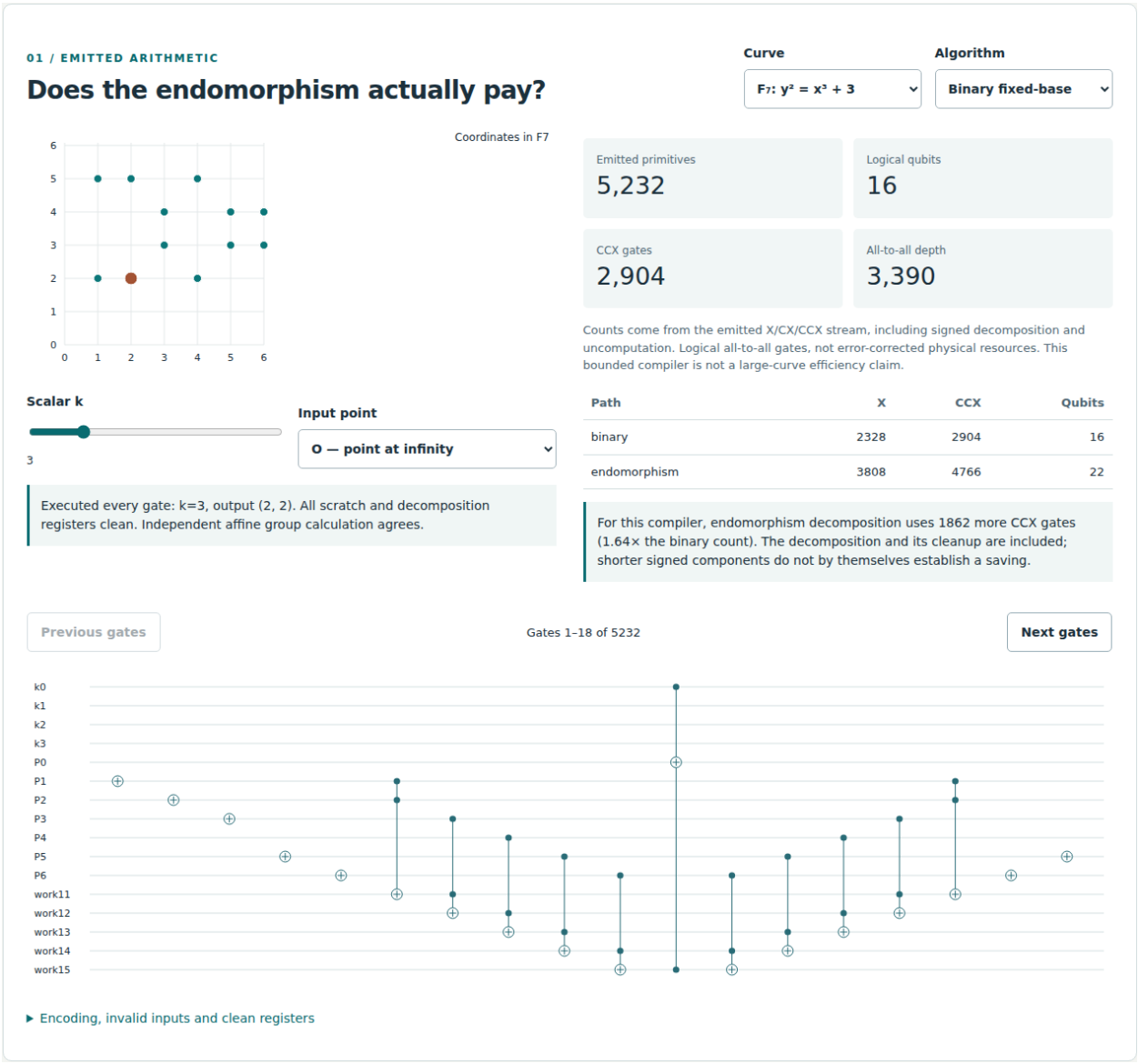


Figure 4.1: The actual arithmetic explorer: finite-field points, complete emitted-gate counts, invalid-word behavior and a paginated circuit stream.

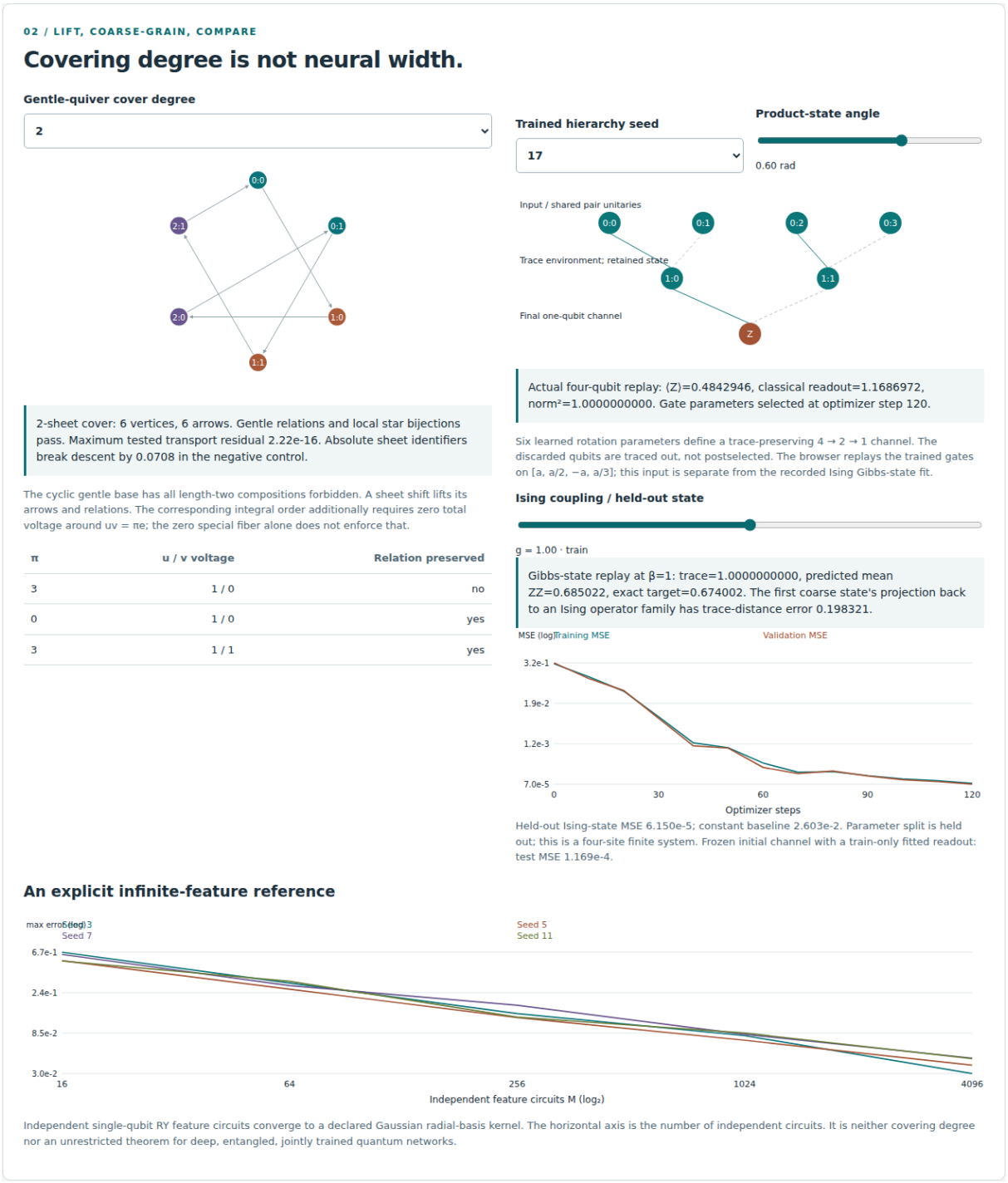


Figure 4.2: Voltage covers, learned hierarchical channels, effective-Hamiltonian error and the separately scaled independent-feature width experiment.

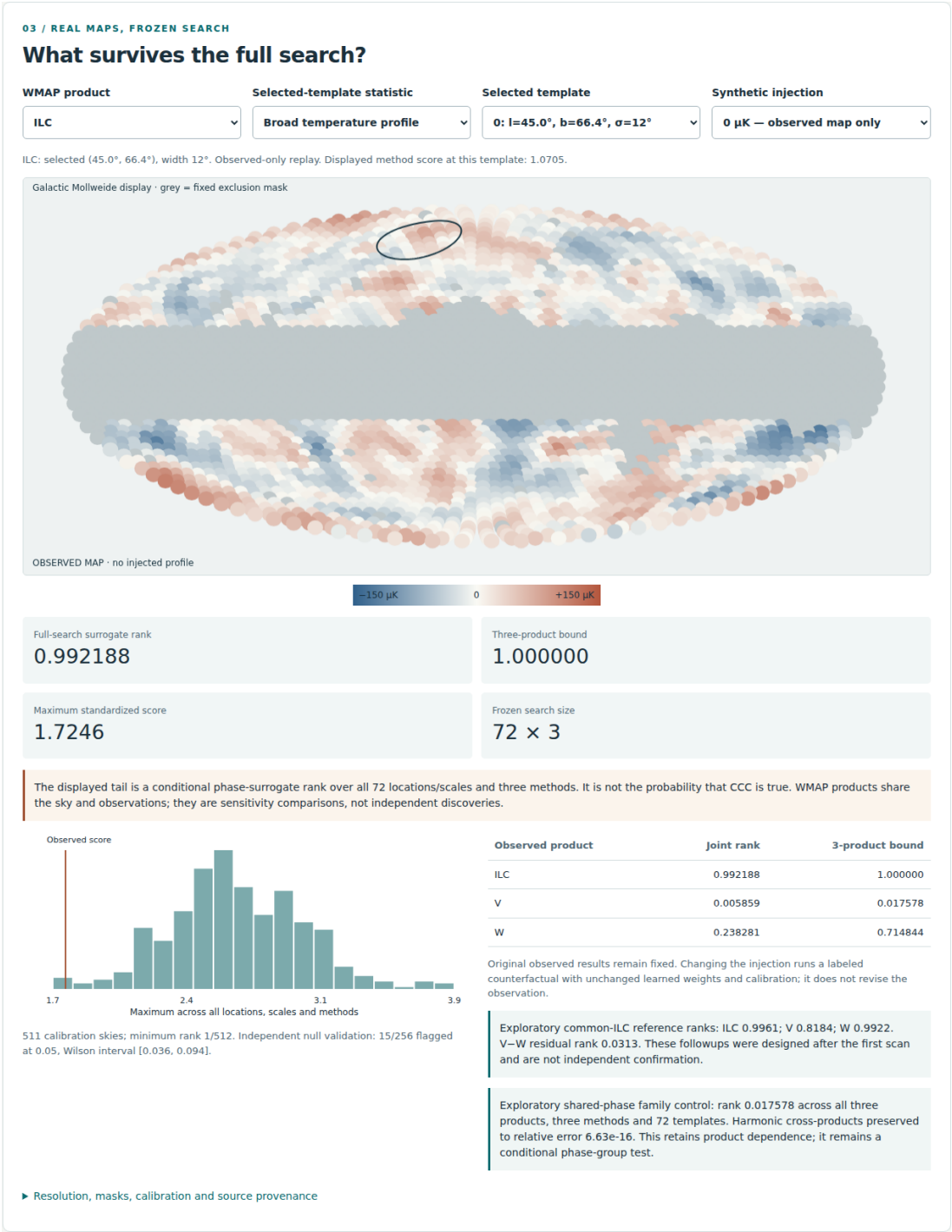


Figure 4.3: Observed WMAP maps and a complete frozen search. Synthetic injections and post-inspection sensitivity tests are explicitly distinguished from the original observations.

Appendix A

Primary sources and evidence boundaries

E1 Roetteler, Naehrig, Svore and Lauter. *Quantum resource estimates for computing elliptic curve discrete logarithms.* ASIACRYPT 2017. https://arxiv.org/abs/1706.06752. This is a scalable circuit-resource reference, not a performance claim for the bounded compiler implemented here.

Q1 Cong, Choi and Lukin. *Quantum Convolutional Neural Networks.* Nature Physics 15, 1273–1278 (2019). https://arxiv.org/abs/1810.03787. The present unitary/trace hierarchy is an explicit finite channel inspired by hierarchical coarse graining; it is not represented as a verbatim implementation of every construction in that paper.

Q2 Girardi and De Palma. *Trained quantum neural networks are Gaussian processes.* Version 2, 2026. https://arxiv.org/abs/2402.08726v2. The light-cone, initialization, kernel and scaling assumptions must accompany any transfer of its trained-network limit.

G1 Assem, Brüstle, Charbonneau-Jodoin and Plamondon. *Gentle algebras arising from surface triangulations.* Algebra and Number Theory 4 (2010), 201–229. https://arxiv.org/abs/0903.3347. The unpunctured marked-surface hypotheses are retained.

G2 De la Peña and Redondo. *Covering functors without groups.* Journal of Algebra 321 (2009), 3816–3826. DOI 10.1016/j.jalgebra.2009.02.023. The finite voltage construction in this volume is specified directly rather than claiming all covering-functor results.

C1 NASA LAMBDA / WMAP Science Team. *WMAP DR5 ILC map description and downloads.* https://lambda.gsfc.nasa.gov/product/wmap/dr5/ilc_map_info.html; https://lambda.gsfc.nasa.gov/product/wmap/dr5/ilc_map_get.html. Product resolution, Galactic coordinates and the approximately ten-degree reliability qualification determine this study's broad scales.

C2 NASA LAMBDA / WMAP Science Team. *WMAP DR5 sky maps and masks.* https://lambda.gsfc.nasa.gov/product/wmap/dr5/skymap_info.html; https://lambda.gsfc.nasa.gov/product/wmap/dr5/masks_get.html. Native filenames, units, ordering and exact bytes are recorded in the acquisition manifest.

C3 The original GRH-research chapter *Cyclic cosmology* and its frozen source ledger. https://github.com/amelie-iska/GRH-research/tree/fd57c5251bc40c388cc25aa83e25f8a2cdf995b7/extension/review/cosmology. Original theoretical claims and published observational comparisons remain separately attributed; this continuation is not a replacement for those source papers.